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TRIPLE-PATH CLOCK AND DATA RECOVERY CIRCUIT, OSCILLATOR CIRCUIT AND METHOD FOR CLOCK AND DATA RECOVERY

Abstract

A clock and data recovery circuit includes a sampling circuit, a phase detector, a first processing circuit, a second processing circuit and an oscillator circuit. The sampling circuit is configured to sample input data according to an output clock, and generate a sampling result. The phase detector is configured to generate a detection result according to the sampling result. The first processing circuit is configured to process the sampling result to generate a first digital code. The second processing circuit is configured to accumulate a portion of the first digital code to generate a second digital code. A rate of change of a code value of the second digital code is slower than a rate of change of a code value of the first digital code. The oscillator circuit is configured to generate the output clock according to the detection result, the first digital code and the second digital code.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] The present application is a continuation application of U.S. patent application Ser. No. 18/429,890 filed on Feb. 1, 2024, which is a divisional application of U.S. patent application Ser. No. 18/146,854 filed on Dec. 27, 2022, which is divisional application of U.S. patent application Ser. No. 17/355,178 filed on Jun. 23, 2021, which claims priority to U.S. Provisional Patent Application No. 63/043,068, filed on Jun. 23, 2020, each of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates to clock and data recovery (CDR) and, more particularly, to a clock and data recovery circuit having a triple-path structure for bandwidth tracking, and a related oscillator circuit and method for clock and data recovery.

[0003] With the use of clock and data recovery (CDR) techniques, a receiver can retrieve data from a data stream transmitted without additional timing information. Firstly, the receiver may perform clock recovery to extract a clock signal embedded in level transitions in the data stream. Next, the receiver can phase-align the clock signal to the level transitions in the data stream, and recover the data from the data stream by sampling the data stream according to the phase-aligned clock signal. For example, CDR circuits have been widely used in high-speed serial interfaces to regenerate a data stream according to a high speed clock signal, which is phase-aligned to level transitions in the data stream. CDR circuits may face several challenges. For example, an oscillator circuit of a CDR circuit may suffer from large jitter resulting from process variations, temperature variations, and/or timing uncertainties in high-speed data transmission.

SUMMARY

[0004] The described embodiments provide a clock and data recovery circuit having a triple-path structure for bandwidth tracking, and a related oscillator circuit and method for clock and data recovery.

[0005] Some embodiments described herein may include a clock and data recovery (CDR) circuit. The CDR circuit includes a sampling circuit, a phase detector, a first processing circuit, a second processing circuit and an oscillator circuit. The sampling circuit is configured to sample input data according to an output clock and accordingly generate a sampling result. The phase detector, coupled to the sampling circuit, is configured to generate a detection result according to the sampling result. The first processing circuit, coupled to the sampling circuit, is configured to process the sampling result to generate a first digital code. The second processing circuit, coupled to the first processing circuit, is configured to accumulate a portion of the first digital code to generate a second digital code. A rate of change of a code value of the second digital code is slower than a rate of change of a code value of the first digital code. The oscillator circuit, coupled to the sampling circuit, the phase detector, the first processing circuit and the second processing circuit, is

configured to generate the output clock according to the detection result, the first digital code and the second digital code. A phase of the output clock is adjusted at least according to the detection result, and a frequency of the output clock is adjusted according to the first digital code and the second digital code.

[0006] Some embodiments described herein may include an oscillator circuit. The oscillator circuit includes a current-controlled oscillator, a first conversion circuit, a second conversion circuit and a third conversion circuit. The current-controlled oscillator is configured to generate an output clock according to a first control current, a second control current and a third control current. A frequency of the output clock is controlled by the first control current and the second control current, and a phase of the output clock is controlled by the third control current. The first conversion circuit, coupled to the current-controlled oscillator, is configured to convert a first digital code to the first control current according to a first reference current. The second conversion circuit, coupled to the current-controlled oscillator and the first conversion circuit, is configured to convert a second digital code to the second control current, the first reference current and a second reference current. The second digital code is an accumulation result of at least one most significant bit of the first digital code. The third conversion circuit, coupled to the current-controlled oscillator and the first conversion circuit, is configured to convert a third digital code and the second reference current to the third control current.

[0007] Some embodiments described herein may include a method for clock and data recovery (CDR). The method includes: generating a data signal and an edge signal by sampling input data according to an output clock outputted from an oscillator, the data signal and the edge signal carrying phase error information on a phase error between the input data and the output clock; generating a detection result according to the data signal and the edge signal, the detection result indicating a phase relationship between the input data and the output clock; accumulating the phase error information carried by the data signal and the edge signal to generate a first digital code; accumulating at least one most significant bit of the first digital code to generate a second digital code; adjusting a phase of the output clock according to the detection result and the second digital code; and adjusting a frequency of the output clock according to the first digital code and the second digital code.

[0008] With the use of the proposed CDR scheme, an oscillator circuit in a CDR circuit can not only have a wide tuning range but also high resolution despite temperature variations. In addition, the proposed CDR scheme can achieve bandwidth tracking at various frequency corners, thereby ensuring good loop stability.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0010] FIG. 1 is a block diagram illustrating an exemplary clock and data recovery circuit in accordance with some embodiments of the present disclosure.

[0011] FIG. 2 illustrates an implementation of the clock and data recovery circuit shown in FIG. 1 in accordance with some embodiments of the present disclosure.

[0012] FIG. 3 illustrates an implementation of the control circuit shown in FIG. 2 in accordance with some embodiments of the present disclosure.

[0013] FIG. 4 illustrates an implementation of the digital-to-analog converter shown in FIG. 3 in accordance with some embodiments of the present disclosure.

[0014] FIG. 5 is a flow chart of an exemplary method for clock and data recovery in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0015] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, parameter values in the description that follows may vary depending on a given technology node. As another example, parameter values for a given technology node may vary depending on a given application or operating scenario. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0016] Further, it will be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

[0017] To ensure loop stability despite temperature variations, a clock and data recovery (CDR) circuit may utilize an oscillator circuit with a wide tuning range. For example, the oscillator circuit may be designed to have a large frequency step to thereby widen a tuning range thereof. However, the large frequency step results in large jitter outputted from the oscillator circuit.

[0018] The present disclosure describes exemplary CDR circuits, each of which can employ a triple-path structure for controlling operation of an oscillator circuit. The oscillator circuit is included in the exemplary CDR circuit, and configured to generate an output clock for recovering data from a data stream. Two circuit paths of the triple-path structure can perform coarse tuning and fine tuning on a frequency of the output clock, respectively. In some embodiments, one of the two circuit paths is configured to receive a digital code to perform the fine tuning, and the other of the two circuit paths is configured to receive an accumulation result of a portion of the digital code to perform the coarse tuning. Additionally, or alternatively, the triple-path structure can achieve bandwidth tracking with the use of the circuit path used for the coarse tuning. The present disclosure further describes exemplary oscillator circuits, each of which can not only have a wide tuning range but also high resolution. Related methods for CDR are also described in the present disclosure. Further description is provided below.

[0019] FIG. 1 is a block diagram illustrating an exemplary CDR circuit in accordance with some embodiments of the present disclosure. The CDR circuit **100** can be configured to recover timing information carried by input data DI.sub.N, such as a data stream, and regenerate the input data D.sub.IN according to the recovered timing information. The CDR circuit **100** includes, but is not limited to, a sampling circuit **110**, a phase detector (PD) **120**, a plurality of processing circuits **130** and **140**, and an oscillator circuit **150**. In the present embodiment, the CDR circuit **100** may employ a triple-loop structure. For example, the sampling circuit **110** may be used together with the PD **120** and the oscillator circuit **150** to implement a phase tracking loop. The sampling circuit **110** may be used together with the processing circuit **130** and the oscillator circuit **150** to implement a frequency tracking loop. Also, the sampling circuit **110** may be used together with the processing circuit **130**, the processing circuit **140** and the oscillator circuit **150** to implement another frequency tracking loop.

[0020] The sampling circuit **110** is configured to sample the input data D.sub.IN according to an output clock CK.sub.OUT, and accordingly generate a sampling result SR. In the present embodiment, the sampling circuit **110** may sample data bits of the input data D.sub.IN and a data edge in between two consecutive data bits according to the output clock CK.sub.OUT, and accordingly generate a data signal DS and an edge signal ES. As a result, the data signal DS and the edge signal ES can carry phase error information on a phase error between the input data D.sub.IN and the output clock CK.sub.OUT. By way of example but not limitation, the sampling circuit **110**

may perform two-times ($2\times$) oversampling on the input data D.sub.IN to generate the sampling result SR.

[0021] The PD **120**, coupled to the sampling circuit **110**, is configured to generate a detection result DR according to the sampling result SR. The detection result DR may indicate whether the output clock CK.sub.OUT lags or leads the input data D.sub.IN. In the present embodiment, the PD **120** may receive the data signal DS and the edge signal ES to detect a difference in phase between the input data D.sub.IN and the output clock CK.sub.OUT. The detection result DR may include an up signal UP and a down signal DN to indicate whether the output clock CK.sub.OUT lags or leads the input data D.sub.IN. For example, the PD **120** may generate the up signal UP with a predetermined logic level when the output clock CK.sub.OUT lags the input data D.sub.IN, and generate the down signal DN with a predetermined logic level when the output clock CK.sub.OUT leads the input data D.sub.IN. In some embodiments, the detection result DR may be implemented using a digital code DC.sub.P which includes the up signal UP and the down signal DN.

[0022] The processing circuit **130**, coupled to the sampling circuit **110**, is configured to process the sampling result SR to generate a digital code DC.sub.I, which can indicate information on a frequency error between the input data D.sub.IN and the output clock CK.sub.OUT. In the present embodiment, the digital code DC.sub.I is implemented as an M-bit digital signal, where M is an integer greater than one.

[0023] The processing circuit **140**, coupled to the processing circuit **130**, is configured to accumulate a portion of the digital code DC.sub.I to generate a digital code DC.sub.F. A rate of change of a code value of the digital code DC.sub.F can be slower than a rate of change of a code value of the digital code DC.sub.I. For example, the portion of the digital code DC.sub.I may be K bits of the digital code DC.sub.I, where K is a positive integer less than M. As another example, the portion of the digital code DC.sub.I may be the most significant bit (MSB) of the digital code DC.sub.I. As another example, the portion of the digital code DC.sub.I may be the first two MSBs of the digital code DC.sub.I. As still another example, the portion of the digital code DC.sub.I may be at least one MSB of the digital code DC.sub.I. In the present embodiment, the digital code DC.sub.F can indicate information on the frequency error between the input data D.sub.IN and the output clock CK.sub.OUT since the digital code DC.sub.F is an accumulation result of the portion of the digital code DC.sub.I. The digital code DC.sub.F can be implemented as an N-bit digital signal, where N is a positive integer.

[0024] The oscillator circuit **150**, coupled to the sampling circuit **110**, the PD **120**, the processing circuit **130** and the processing circuit **140**, is configured to generate the output clock CK.sub.OUT according to the detection result DR, the digital code DC.sub.I and the digital code DC.sub.F. A phase of the output clock CK.sub.OUT is adjusted at least according to the detection result DR, and a frequency of the output clock CK.sub.OUT is adjusted according to the digital code DC.sub.I and the digital code DC.sub.F. In some embodiments where the detection result DR is implemented using the digital code DC.sub.P, the oscillator circuit **150** can be implemented as a digitally-controlled oscillator (DCO) circuit. Additionally, or alternatively, as the rate of change of the code value of the digital code DC.sub.F can be slower than that of the code value of the digital code DC.sub.I, the digital code DC.sub.F can be regarded as a control input for coarse tuning of the frequency of the output clock CK.sub.OUT, and the digital code DC.sub.I can be regarded as a control input for fine tuning of the frequency of the output clock CK.sub.OUT.

[0025] In the present embodiment, the oscillator circuit **150** includes, but is not limited to, a control circuit **160** and an oscillator **170**. The control circuit **160**, coupled to the PD **120**, the processing circuit **130** and the processing circuit **140**, may utilize a triple-path structure to control operation of the oscillator **170**. For example, the control circuit **160** is configured to generate a control signal CS.sub.P at least according to the detection result DR, generate a control signal CS.sub.I at least according to the digital code DC.sub.I, and generate a control signal CS.sub.F according to the digital code DC.sub.F. The control signal CS.sub.P, generated from one path in the control circuit

160, is sent to the oscillator **170** to adjust the phase of the output clock CK.sub.OUT. The control signals CS.sub.I and CS.sub.F, generated from other two paths in the control circuit **160**, are sent to the oscillator **170** to adjust the frequency of the output clock CK.sub.OUT. An increase in a signal level of the control signal CS.sub.F, generated when the code value of the digital code DC.sub.F is incremented by a predetermined amount such as a binary 1, is greater than an increase in a signal level in the control signal CS.sub.I, generated when the code value of the digital code DC.sub.I is incremented by the predetermined amount. As a result, the control signal CS.sub.F can be used for coarse tuning of the frequency of the output clock CK.sub.OUT, and the control signal CS.sub.I can be used for fine tuning of the frequency of the output clock CK.sub.OUT.

[0026] The oscillator **170**, coupled to the sampling circuit **110** and the control circuit **160**, is configured to generate the output clock CK.sub.OUT according to the control signal CS.sub.P, the control signal CS.sub.I and the control signal CS.sub.F. The oscillator **170** may be implemented using a current-controlled oscillator (CCO), a voltage-controlled oscillator (VCO) or a hybrid current/voltage-controlled oscillator.

[0027] In operation, the sampling circuit **110** may oversample the input data D.sub.IN according to the output clock CK.sub.OUT, and accordingly generate the data signal DS and the edge signal ES. The PD **120** may output the up signal UP and the down signal DN according to whether the output clock CK.sub.OUT lags or leads the input data D.sub.IN. The control circuit **160** may utilize the up signal UP and the down signal DN to generate the control signal CS.sub.P to thereby adjust the phase of the output clock CK.sub.OUT. A circuit path associated with generation of the control signal CS.sub.P can be referred to as a proportional path (denoted as P-path). The processing circuit **130** may process the sampling result SR to accumulate the phase error information carried by the data signal DS and the edge signal ES, thereby producing the digital code DC.sub.I which indicates the information on the frequency error between the input data D.sub.IN and the output clock CK.sub.OUT. The control circuit **160** may utilize the digital code DC.sub.I to generate the control signal CS.sub.I to thereby adjust the frequency of the output clock CK.sub.OUT. A circuit path associated with generation of the control signal CS.sub.I can be referred to as an integral path (denoted as I-path). In addition, the processing circuit **140** may accumulate at least one MSB of the digital code DC.sub.I to produce the digital code DC.sub.F. The control circuit **160** may utilize the digital code DC.sub.F to generate the control signal CS.sub.F to thereby adjust the frequency of the output clock CK.sub.OUT. A circuit path associated with generation of the control signal CS.sub.F is denoted as F-path.

[0028] As the code value of the digital code DC.sub.F changes more slowly than the code value of the digital code DC.sub.I, the control circuit **160** may utilize the digital code DC.sub.F to coarse tune the frequency of the output clock CK.sub.OUT, and utilize the digital code DC.sub.I to fine tune the frequency of the output clock CK.sub.OUT. For example, when the code value of the digital code DC.sub.F is unchanged and the code value of the digital code DC.sub.I increases, the signal level of the control signal CS.sub.I increases. The frequency of the output clock CK.sub.OUT can be adjusted using a fine step size. When the code value of the digital code DC.sub.F increases, the signal level of the control signal CS.sub.F increases. The frequency of the output clock CK.sub.OUT can be adjusted using a coarse step size.

[0029] It is worth noting that, with the use of the proposed CDR scheme, the gain associated with the integral path of the control circuit **160** can be kept small to provide high frequency resolution. In addition, the coarse tuning implemented using the processing circuit **140** can provide a large frequency step size and therefore allow the oscillator circuit **150** to have a wide tuning range. Moreover, the control circuit **160** may generate the control signal CS.sub.P according to the detection result DR and the digital code DC.sub.F. By adjusting respective signal levels of the control signals CS.sub.P, CS.sub.I and CS.sub.F according to the digital code DC.sub.F, the control circuit **160** can allow each of the control signal CS.sub.P generated from the P-path and the control signal CS.sub.I generated from the I-path to track the control signal CS.sub.F generated from the F-

path, thereby achieving bandwidth tracking at various frequency corners.

[0030] In some embodiments, before the CDR circuit **100** starts to track the input data DI.sub.N, the processing circuit **140** can be configured to compare a frequency of a reference clock CK.sub.R with the frequency of the output clock CK.sub.OUT, and accordingly set the code value of the digital code DC.sub.F to a predetermined value. When the CDR circuit **100** starts to track the input data DI.sub.N, the processing circuit **140** can accumulate the portion of the digital code DC.sub.I to update the code value of the digital code DC.sub.F. With the use of the predetermined value, the CDR circuit **100** can shorten a period of time it takes to lock the output clock CK.sub.OUT.

[0031] To facilitate understanding of the present disclosure, some embodiments are given as follows for further description of the proposed CDR scheme. Those skilled in the art should appreciate that other embodiments employing the architecture shown in FIG. **1** are also within the contemplated scope of the present disclosure.

[0032] FIG. **2** illustrates an implementation of the CDR circuit **100** shown in FIG. **1** in accordance with some embodiments of the present disclosure. The CDR circuit **200** includes, but is not limited to, the sampling circuit **110** and the PD **120** shown in FIG. **1**, a plurality of processing circuits **230** and **240**, and an oscillator circuit **250**. The processing circuits **230** and **240** can represent embodiments of the processing circuits **130** and **140** shown in FIG. **1**, respectively. The oscillator circuit **250** can represent an embodiment of the oscillator circuit **150** shown in FIG. **1**.

[0033] The processing circuit **230** includes, but is not limited to, a deserializer **232** and an accumulator **236**. The deserializer **232**, coupled to the sampling circuit **110**, is configured to process the data signal DS and the edge signal ES to generate a deserialization result DES. The deserialization result DES can indicate the phase error information on the phase error between the input data D.sub.IN and the output clock CK.sub.OUT. The accumulator **236**, coupled to the deserializer **232**, is configured to accumulate the phase error information indicated by the deserialization result DES to generate the digital code DC.sub.I.

[0034] The processing circuit **240** includes, but is not limited to, a calibration circuit **242** and an accumulator **246**. The calibration circuit **242**, coupled to the oscillator circuit **250**, is configured to compare the frequency of the reference clock CK.sub.R with the frequency of the output clock CK.sub.OUT to generate a calibration result CR. The accumulator **246**, coupled to the calibration circuit **242**, is configured to set the code value of the digital code DC.sub.F according to the calibration result CR, and accumulate a portion of the digital code DC.sub.I to update the code value of the digital code DC.sub.F after the code value of the digital code DC.sub.F is set according to the calibration result CR.

[0035] The oscillator circuit **250** includes, but is not limited to, a control circuit **260** and a current-controlled oscillator (CCO) **270**. The control circuit **260** is configured to generate a control current I.sub.I at least according to the digital code DC.sub.I, generate a control current I.sub.F according to the digital code DC.sub.F, and generate a control current I.sub.P at least according to the digital code DC.sub.P (i.e. the detection result DR outputted from the PD **120**). The control currents I.sub.P, I.sub.I and I.sub.F can represent embodiments of the control signals CS.sub.P, CS.sub.I and CS.sub.F shown in FIG. **1**, respectively. An increment in the control current I.sub.F, generated when the code value of the digital code DC.sub.F is incremented by a predetermined amount such as binary 1, is greater than an increment in the control current II, generated when the code value of the digital code DC.sub.I is incremented by the predetermined amount. As a result, the control current I.sub.F can be used for coarse tuning of the frequency of the output clock CK.sub.OUT, and the control current I.sub.I can be used for fine tuning of the frequency of the output clock CK.sub.OUT.

[0036] In the present embodiment, the control circuit **260** can be configured to generate the control current I.sub.I according to the digital code DC.sub.I and the digital code DC.sub.F, and generate the control current I.sub.P according to the digital code DC.sub.P and the digital code DC.sub.F. As a result, the control circuit **260** can allow each of the control current I.sub.P and the control current

I.sub.I to track the control current I.sub.F, thus achieving bandwidth tracking and ensure good loop stability. For example, the control circuit **260** may include a plurality of conversion circuits **262**, **264** and **266**, which can be used to implement the P-path, the I-path and the F-path shown in FIG. **1**, respectively. The control circuit **260** may allow at least one of the control current I.sub.I generated from the conversion circuit **262** and the control current I.sub.P generated from the conversion circuit **266** to track the control current IF generated from the conversion circuit **264**. [0037] The conversion circuit **262**, coupled to the processing circuit **230**, is configured to generate the control current I.sub.I according to the digital code DC.sub.I and a reference current I.sub.REFI. For example, the conversion circuit **262** is configured to convert the digital code DC.sub.I to the control current I.sub.I according to the reference current I.sub.REFI. A current level of the control current I.sub.I changes in response to a current level of the reference current I.sub.REFI when the code value of the digital code DC.sub.I is kept unchanged.

[0038] The conversion circuit **264**, coupled to the processing circuit **240** and the conversion circuit **262**, is configured to generate the control current I.sub.F and the reference current I.sub.REFI according to the digital code DC.sub.F. For example, the conversion circuit **264** is configured to convert the digital code DC.sub.F, i.e. an accumulation result of a portion of the digital code DC.sub.I, to the control current I.sub.F and the reference current I.sub.REFI. When the code value of the digital code DC.sub.F increases, each of the control current I.sub.F and the reference current I.sub.REFI may increase. In the present embodiment, the conversion circuit **264** is further configured to convert the digital code DC.sub.F to a reference current I.sub.REFP. Respective current levels of the reference currents I.sub.REFI and I.sub.REFP can change in response to a current level of the control current I.sub.F when the code value of the digital code DC.sub.F is kept unchanged. In other words, each of the reference currents I.sub.REFI and I.sub.REFP, coming from the conversion circuit **264**, can track the control current I.sub.F. For example, when the code value of the digital code DC.sub.F increases, each of the reference current I.sub.REFI and the reference current I.sub.REFP may increase.

[0039] The conversion circuit **266**, coupled to the PD **120** and the conversion circuit **264**, is configured to convert the digital code DC.sub.P and the reference current I.sub.REFP to the control current I.sub.P. For example, the conversion circuit **266** can be configured to generate the control current I.sub.P by selectively steering the reference current I.sub.REFP from the conversion circuit **264** to the CCO **270** according to the digital code DC.sub.P. A current level of the control current I.sub.P changes in response to a current level of the reference current I.sub.REFP when the code value of the digital code DC.sub.P is kept unchanged.

[0040] The CCO **270**, coupled to the conversions circuit **262**, **264** and **266**, is configured to generate the output clock CK.sub.OUT according to the control currents I.sub.P, I.sub.I and I.sub.F. In the present embodiment, the frequency of the output clock CK.sub.OUT is controlled by the control current I.sub.I and the control current I.sub.F, and the phase of the output clock CK.sub.OUT is controlled by the control current I.sub.P.

[0041] In operation, before the CDR circuit **200** starts to track the input data DI.sub.N, the calibration circuit **242** may be activated to generate the calibration result CR by comparing the frequency of the reference clock CK.sub.R with the frequency of the output clock CK.sub.OUT. The accumulator **246** may set the code value of the digital code DC.sub.F to a predetermined value according to the calibration result CR. After the CDR circuit **200** starts to track the input data DI.sub.N, the calibration circuit **242** may be deactivated. The sampling circuit **110** may oversample the input data D.sub.IN according to the output clock CK.sub.OUT, and accordingly generate the data signal DS and the edge signal ES.

[0042] With regard to frequency tracking loops, the deserializer **232** may convert the data signal DS and the edge signal ES in serial form into the deserialization result DES which is in parallel form. The accumulator **236** may accumulate the phase error information indicated by the deserialization result DES to produce the digital code DC.sub.I, which indicates the information on

the frequency error between the input data D.sub.IN and the output clock CK.sub.OUT. The accumulator **246** may accumulate the portion of the digital code DC.sub.I from the predetermined value to update the code value of the digital code DC.sub.F.

[0043] In addition, the conversion circuit **264** may generate the control current I.sub.F and the reference current I.sub.REFI according to the digital code DC.sub.F. The conversion circuit **262** may convert the digital code DC.sub.I to the control current I.sub.I according to the reference current I.sub.REFI provided by the conversion circuit **264**. When the code value of the digital code DC.sub.F increases, the control current I.sub.F increases. When a code value of another portion of the digital code DC.sub.I increases, the control current I increases. For example, the accumulator **246** updates the digital code DC.sub.F by accumulating at least one MSB of the digital code DC.sub.I from the predetermined value. The another portion of the digital code DC.sub.I may be at least one least significant bit (LSB) of the digital code DC.sub.I which is not accumulated by the accumulation **246**. When the code value of the digital code DC.sub.F is unchanged, and the code value of the another portion of the digital code DC.sub.I increases, it means that the control circuit **260** fine tunes the frequency of the output clock CK.sub.OUT. The control current I.sub.I increases in response to the code value of the digital code DC.sub.I. The frequency of the output clock CK.sub.OUT can be adjusted using a fine step size. When the code value of the digital code DC.sub.F changes or increases, it means that the control circuit **260** coarse tunes the frequency of the output clock CK.sub.OUT. The control current I.sub.F may increase accordingly. The frequency of the output clock CK.sub.OUT can be adjusted using a coarse step size.

[0044] With regard to a phase tracking loop, the PD **120** may output the up signal UP and the down signal DN according to the data signal DS and the edge signal ES. The conversion circuit **266** may convert the digital code DC.sub.P and the reference current I.sub.REFP, provided by the conversion circuit **264**, to the control current I.sub.P. For example, when the digital code DC.sub.P has a predetermined code value, the conversion circuit **266** is configured to increase the control current I.sub.P by steering the reference current I.sub.REFP from the conversion circuit **264** to the CCO **270**. When the digital code DC.sub.P has another predetermined code value, the conversion circuit **266** is configured to decrease the control current I.sub.P by stopping steering the reference current I.sub.REFP to the CCO **270**. The phase of the output clock CK.sub.OUT can be adjusted according to the control current I.sub.P.

[0045] The above circuit structures are provided for illustrative purposes, and are not intended to limit the scope of the present disclosure. In some embodiments, the calibration circuit **242** may be optional. In some embodiments, the oscillator circuit **250** may be implemented using a VCO circuit or a hybrid current/voltage-controlled oscillator circuit. Such modifications and alternatives also fall within the contemplated scope of the present disclosure.

[0046] FIG. **3** illustrates an implementation of the control circuit **260** shown in FIG. **2** in accordance with some embodiments of the present disclosure. The control circuit **360** includes a plurality of conversion circuits **362**, **364** and **366**, which represent embodiments of the conversion circuits **262**, **264** and **266** shown in FIG. **2**, respectively. In the present embodiment, the conversion circuit **362** includes, but is not limited to, a current mirror circuit **372**, a digital-to-analog converter (DAC) **382** and a voltage-to-current converter **392**. The current mirror circuit **372** can be configured to mirror the reference current I.sub.REFI to generate an auxiliary current I.sub.AUXI. By way of example but not limitation, the current mirror circuit **372** may include a plurality of transistors M31 and M32. The DAC **382**, coupled to the current mirror circuit **372**, can be configured to convert the digital code DC.sub.I to an auxiliary voltage V.sub.AUXI according to the auxiliary current I.sub.AUXI. The voltage-to-current converter **392**, coupled to the DAC **382**, can be configured to convert the auxiliary voltage V.sub.AUXI to the control current I_I. By way of example but not limitation, the voltage-to-current converter **392** may include an amplifier A1, a resistor R1 and a plurality of transistors M33 and M34.

[0047] The conversion circuit **364** includes, but is not limited to, a DAC **374** and a voltage-to-

current converter **384**. The DAC **374** can be configured to convert the digital code DC.sub.F to an auxiliary voltage V.sub.AUXF. The voltage-to-current converter **384**, coupled to the DAC **374**, can be configured to convert the auxiliary voltage VAUXF to the reference current I.sub.REFI, the reference current I.sub.REFP and the control current I.sub.F. By way of example but not limitation, the voltage-to-current converter **384** may include an amplifier A2, a resistor R2 and a plurality of transistors M35-M39.

[0048] The conversion circuit **366** includes, but is not limited to, a plurality of switches SW.sub.U and SW.sub.D. The switches SW.sub.U and SW.sub.D can be controlled by the up signal UP and the down signal DN provided by the PD **120** shown in FIG. 2, respectively. As those skilled in the art can appreciate the generation of the control currents I.sub.P, I.sub.I and I.sub.F after reading the above paragraphs directed to FIG. 1 and FIG. 2, further description is omitted here for brevity.

[0049] FIG. 4 illustrates an implementation of the DAC **382** shown in FIG. 3 in accordance with some embodiments of the present disclosure. The DAC **382** is configured to convert the digital code DC.sub.I, implemented as an M-bit digital signal, to the auxiliary voltage V.sub.AUXI according to the auxiliary current I.sub.AUXI. In the present embodiment, the DAC **382** includes a plurality of transistors M4.sub.0-M4.sub.M+1, a plurality of switches SW.sub.0-SW.sub.M, and a resistor R4. The switches SW.sub.0-SW.sub.M are controlled by M bits B.sub.0-B.sub.M of the digital code DC.sub.I. As a result, a voltage level of the auxiliary voltage V.sub.AUXI can be determined according to the number of switches which are turned on. In some embodiments, the DAC **374** shown in FIG. 3 may employ a circuit structure similar to that of the DAC **382** shown in FIG. 4.

[0050] It is worth noting that the above circuit implementations shown in FIG. 3 and FIG. 4 are provided for illustrative purposes, and are not intended to limit the scope of the present disclosure. As long as a control circuit employs a triple-path structure, which accumulates a digital code for fine frequency tuning and accumulates a portion of the digital code for coarse frequency tuning, to control operation of an oscillator, associated modifications and alternatives fall within the contemplated scope of the present disclosure. Additionally, or alternatively, as long as a control circuit employs a triple-path structure, which allows respective control signals provided by a proportional path and an integral path to track a control signal provided by another path used for coarse frequency tuning, to control operation of an oscillator, associated modifications and alternatives fall within the contemplated scope of the present disclosure.

[0051] FIG. 5 is a flow chart of an exemplary method for clock and data recovery (CDR) in accordance with some embodiments of the present disclosure. The method **500** is described with reference to the CDR circuit **200** shown in FIG. 2 for illustrative purposes. Those skilled in the art should appreciate that the method **500** can be employed in the CDR circuit **100** shown in FIG. 1 or other CDR circuits having triple-path structure without departing from the scope of the present disclosure. Additionally, in some embodiments, other operations in the method **500** can be performed. In some embodiments, operations of the method **500** can be performed in a different order and/or vary. In some other embodiments, one or more operations of the method **500** may be optional.

[0052] At operation **502**, a data signal and an edge signal are generated by sampling input data according to an output clock outputted from an oscillator. The data signal and the edge signal carry phase error information on a phase error between the input data and the output clock. For example, the sampling circuit **110** can generate the data signal DS and edge signal ES by sampling the input data D.sub.IN according to the output clock CK.sub.OUT outputted from the CCO **270**. The data signal DS and the edge signal ES can carry phase error information on a phase error between the input data D.sub.IN and the output clock CK.sub.OUT.

[0053] At operation **504**, a detection result is generated according to the data signal and the edge signal. The detection result indicates a phase relationship between the input data and the output clock. For example, the PD **120** can generate the detection result DR according to the data signal

DS and edge signal ES, wherein the detection result DR, including the up signal UP and the down signal DN, can indicate a phase relationship between the input data D.sub.IN and the output clock CK.sub.OUT.

[0054] At operation **506**, the phase error information, carried by the data signal and the edge signal, is accumulated to generate a first digital code. For example, the deserializer **232** can process the data signal DS and the edge signal ES in serial form to generate the deserialization result DES in parallel form, wherein the deserialization result DES can indicate the phase error information on the phase error between the input data D.sub.IN and the output clock CK.sub.OUT. The accumulator **236** can accumulate the phase error information indicated by the deserialization result DES to generate the digital code DC.sub.I.

[0055] At operation **508**, at least one MSB of the first digital code is accumulated to generate a second digital code. For example, the accumulator **246** can accumulate one or more MSBs of the digital code DC.sub.I to generate the digital code DC.sub.F.

[0056] At operation **510**, a phase of the output clock is adjusted according to the detection result and the second digital code. For example, the control circuit **260** can adjust the phase of the output clock CK.sub.OUT according to the detection result DR and the digital code DC.sub.F.

[0057] At operation **512**, a frequency of the output clock is adjusted according to the first digital code and the second digital code. For example, the control circuit **260** can adjust the frequency of the output clock CK.sub.OUT according to the digital code DC.sub.I and the digital code DC.sub.F. It is worth noting that as the phase and the frequency of the output clock CK.sub.OUT can be adjusted according to the digital code DC.sub.F, the CDR circuit **200** can achieve bandwidth tracking and provide good loop stability.

[0058] In some embodiment, at operation **508**, an initial value of the second digital code can be determined according to a reference clock and the output clock, which can shorten a period of time it takes to lock the output clock. For example, before the CDR circuit **200** starts to track the input data DI.sub.N, the calibration circuit **242** may compare the frequency of the reference clock CK.sub.R with the frequency of the output clock CK.sub.OUT to generate the calibration result CR. Next, the accumulator **246** may set the code value of the digital code DC.sub.F according to the calibration result CR. After the code value of the digital code DC.sub.F is set according to the calibration result CR, the accumulator **246** may update the code value of the digital code DC.sub.F by accumulating the one or more MSBs of the digital code DC.sub.I.

[0059] As those skilled in the art can appreciate operation of the method **500** after reading the above paragraphs directed to FIG. **1** through FIG. **4**, further description is omitted here for brevity.

[0060] With the use of the proposed CDR scheme, an oscillator circuit in a CDR circuit can not only have a wide tuning range but also high resolution despite temperature variations. In addition, the proposed CDR scheme can achieve bandwidth tracking at various frequency corners, thereby ensuring good loop stability.

[0061] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An oscillator circuit, comprising: an oscillator, configured to generate an output clock; a control circuit, coupled to the oscillator, the control circuit being configured to output a first control signal

and a second control signal to the oscillator, and accordingly adjust a frequency of the output clock of the oscillator, wherein the control circuit comprises: a first circuit path, configured to receive a first digital code to generate the first control signal; and a second circuit path, configured to receive a second digital code to generate the second control signal, wherein the second digital code is an accumulation result of a portion of the first digital code; when a code value of first digital code changes, the frequency of the output clock is adjusted using a first step size; when the code value of first digital code is unchanged and a code value of the second digital code changes, the frequency of the output clock is adjusted using a second step size smaller than the first step size.

2. The oscillator circuit of claim 1, wherein the portion of the first digital code is at least one most significant bit of the first digital code.

3. The oscillator circuit of claim 1, wherein an increment in the second control signal, generated when the code value of the second digital code is incremented by a predetermined amount, is greater than an increment in the first control signal, generated when the code value of the first digital code is incremented by the predetermined amount.

4. The oscillator circuit of claim 1, wherein the control circuit is further configured to output a third control signal to the oscillator to adjust a phase of the output clock of the oscillator; the control circuit further comprises: a third circuit path, configured to receive a third digital code to generate the third control signal, wherein respective signal levels of the first control signal, the second control signal and the third control signal are adjusted according to the second digital code.

5. The oscillator circuit of claim 1, wherein the first circuit path is configured to generate a first control current according to the first digital code and a first reference current, and the first control current serves as the first control signal; the second circuit path is configured to generate a second control current and the first reference current according to the second digital code, and the second control current serves as the second control signal.

6. The oscillator circuit of claim 5, wherein the second circuit path is configured to generate a second reference current according to the second digital code; the control circuit further comprises: a third circuit path, configured to generate a third control current by selectively steering the second reference current from the second circuit path to the oscillator according to the third digital code, wherein a phase of the output clock of the oscillator is controlled by the third control current.

7. The oscillator circuit of claim 6, wherein when the code value of the second digital code increases, each of the first reference current and the second reference current increases.

8. The oscillator circuit of claim 5, wherein the first circuit path comprises: a current mirror circuit, configured to mirror the first reference current to generate an auxiliary current; a digital-to-analog converter, coupled to the current mirror circuit, the digital-to-analog converter being configured to convert the first digital code to an auxiliary voltage according to the auxiliary current; and a voltage-to-current converter, coupled to the digital-to-analog converter, the voltage-to-current converter being configured to convert the auxiliary voltage to the first control current.

9. The oscillator circuit of claim 5, wherein the second circuit path comprises: a digital-to-analog converter, configured to convert the second digital code to an auxiliary voltage; and a voltage-to-current converter, coupled to the digital-to-analog converter, the voltage-to-current converter being configured to convert the auxiliary voltage to the first reference current and the second control current.

10. A clock signal generating circuit, comprising: an oscillator, configured to generate an output clock according to a first control signal, a second control signal and a third control signal, wherein a frequency of the output clock is controlled by the first control signal and the second control signal, and a phase of the output clock is controlled by the third control signal; a control circuit, coupled to the oscillator and the processing circuit, the control circuit being configured to generate the first control signal at least according to a first digital code, generate the second control signal according to a second digital code, generate the third control signal at least according to a third digital code, and output the first control signal, the second control signal and the third control

signal to the oscillator to lock the output clock; and a processing circuit, configured to accumulate a portion of the first digital code to generate the second digital code, wherein when a code value of first digital code changes, the frequency of the output clock is adjusted using a first step size; when the code value of first digital code is unchanged and a code value of the second digital code changes, the frequency of the output clock is adjusted using a second step size smaller than the first step size.

11. The clock signal generating circuit of claim 10, wherein the portion of the first digital code is at least one most significant bit of the first digital code.

12. The clock signal generating circuit of claim 10, wherein an increment in the second control signal, generated when the code value of the second digital code is incremented by a predetermined amount, is greater than an increment in the first control signal, generated when the code value of the first digital code is incremented by the predetermined amount.

13. The clock signal generating circuit of claim 10, wherein the control circuit comprises: a first circuit path, configured to receive the first digital code to generate the first control signal; a second circuit path, configured to receive the second digital code to generate the second control signal; and a third circuit path, configured to receive the third digital code to generate the third control signal, wherein respective signal levels of the first control signal, the second control signal and the third control signal are adjusted according to the second digital code.

14. The clock signal generating circuit of claim 13, wherein the first circuit path is configured to generate a first control current according to the first digital code and a first reference current, and the first control current serves as the first control signal; the second circuit path is configured to generate a second control current and the first reference current according to the second digital code, and the second control current serves as the second control signal.

15. The clock signal generating circuit of claim 14, wherein the second circuit path is configured to generate a second reference current according to the second digital code; the third circuit path is configured to selectively steer the second reference current to the oscillator through the second circuit path according to the third digital code, and accordingly generate a third control current serving as the third control signal.

16. The clock signal generating circuit of claim 15, wherein when the code value of the second digital code increases, each of the first reference current and the second reference current increases.

17. The clock signal generating circuit of claim 14, wherein the first circuit path comprises: a current mirror circuit, configured to mirror the first reference current to generate an auxiliary current; a digital-to-analog converter, coupled to the current mirror circuit, the digital-to-analog converter being configured to convert the first digital code to an auxiliary voltage according to the auxiliary current; and a voltage-to-current converter, coupled to the digital-to-analog converter, the voltage-to-current converter being configured to convert the auxiliary voltage to the first control current.

18. The clock signal generating circuit of claim 14, wherein the second circuit path comprises: a digital-to-analog converter, configured to convert the second digital code to an auxiliary voltage; and a voltage-to-current converter, coupled to the digital-to-analog converter, the voltage-to-current converter being configured to convert the auxiliary voltage to the first reference current and the second control current.

19. The clock signal generating circuit of claim 10, wherein the processing circuit is configured to set the code value of the second digital code to a predetermined value by comparing a frequency of a reference clock with the frequency of the output clock, and accumulate the portion of the first digital code to update the code value of the second digital code after the code value of the second digital code is set to the predetermined value.

20. A method for locking an output clock of an oscillator, comprising: accumulating a portion of a first digital code to generate a second digital code; generating a first control current according to the first digital code and the second digital code; generating a second control current according to

the second digital code; generating a third control current according to the second digital code and a third digital code, wherein each of the first control current and the third control current tracks the second control current; adjusting a frequency of the output clock according to the first control current and the second control current, wherein when a code value of first digital code changes, the frequency of the output clock is adjusted using a first step size; when the code value of first digital code is unchanged and a code value of the second digital code changes, the frequency of the output clock is adjusted using a second step size smaller than the first step size; and adjusting a phase of the output clock according to the third control current.
